

Software engineering intro

Condensed Study Notes

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Course Contents

Introduction to Software Design Concepts\n

You'll be exploring core software design aspects such as architecture, patterns, and the use of APIs.

The foundational areas we will investigate include requirements analysis and modeling, application testing (including tool integration), and design for reusability. We'll look at how to leverage APIs through coding.

Having a clear understanding is key to creating robust software.

Software – The Core Concept

At its core, software is coded instructions designed to accomplish tasks for users.

It encompasses not only the code itself but also supporting materials like manuals and documentation that guide users through a system.

The overall objective of software systems is to provide specific functionalities; these are achieved using a set of instructions. These rules affect how data operations can be conducted, or converted into a usable format for an individual's use.

*Think of it as a robot following directions – the instructions give it the capability to perform specific actions but isn't automatically 'alive'.

Software is usually implemented as distinct modules which work in collaboration with each other

Today's software is comprised of data structures and system functionalities

Software systems are built with structured data to facilitate efficient operations. These include essential data structures.

The core structure comprises these fundamental elements:

- Data Structures: Collections of information organized in a particular way that enables programs to operate efficiently (e.g., linked lists, arrays, hash tables)
- Instructions (Computer Programs): Discrete sets of commands that the software implements and carries out by executing instructions.
- Documents: Desired plans for software which are detailed designs including inputs, outputs and UML diagrams
- Installation: Installation involves deploying and launching the software in a platform system; this is important to ensure functionality on user hardware.

The core system functions include executing those processes defined by written programming.

These tools often rely on the specific operating systems/hardware platforms that define its capability with the target hardware architecture (e.g. Windows, macOS)

7. Implementation: Software development must be applied and implemented for different platforms at various stages and functions in a process called installation using several different methodologies to install them properly.

The Term Engineering

// This section defines 'engineering' - focusing on application, control, and design.

The core of engineering is precisely crafting solutions to problems through thoughtful planning and construction.

Engineering is* the act of designing effective systems—whether it's building roads, machines, or software — in a way that achieves specific goals. This necessitates understanding constraints.

Mnemonic: Engineering strives for control; measurable results translate into designed outcomes."

Understanding Visual Basic Source Code vs. Executable Software

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Source code is human-readable, using language constructs for instructions – it's essentially written in plain English.

The executable (or compiled program) is the product of a computer using a language like VB. It's what actually *runs* because the machine understands and carries out each instruction.

Visual Basic (VB), however, is very tightly coupled to a specific BASIC version. Unlike source code, which can be edited easily without changing its fundamental structure, compiled executable code is a rigid translation of those instructions into machine-readable formats – it's treated like an object that cannot be readily modified.

Minimally this means because VB code runs internally and its output does not directly affect the environment; it's created using pre-defined languages or instruction sets which make the code inherently immutable.

Think of it as sending a blueprint to build something – source code is like giving instructions explicitly, and executable software is assembled after the blueprint is refined during compilation.

This crucial distinction means VB isn't easily adaptable. You have little say over how the code behaves as it executes; the interpreter's role is primarily execution – very little modification occurs because of source code.

It also doesn't handle multiple hardware units, directly or implicitly outside its core function.

The Term Software Engineering

(**Section summary:** Software engineering is a disciplined method applied to building and maintaining software, focusing on achieving practical outcomes through careful design.

- **Key Elements:** It centers around analyzing user needs (through product-centric designs) and creating functional applications using programming languages. These applications must fulfill the requirements defined by users for optimal success.
- **Engineering Approach:** The process involves rigorous problem-solving, prioritizing best practices to guarantee robust development and support of the software as a whole.

Computer Science VS Software Engineering

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The following explanation differentiates between Computer Science and Software Engineering, two significant areas within computer science."

"Computer Science's focus is centered on the theoretical foundations and mathematical underpinnings of computers – everything from algorithms to the physical structure. Think of it as understanding *how* a computer works at the core level."

The key requirements are analogous to complex engineering design, involving planning, resource allocation, testing -- all essential for creation."

Rigorously analyzing code is as much about problem solving and creating practical solutions via clever programming techniques"

"Software Engineering emphasizes building *working* applications – ensuring these systems interact correctly and reliably. The goal is that an application will solve user problems with an intuitive process."

OR

Definition & Core Concept

Raw content suggests software engineering aims to systematically address and manage the creation of software, emphasizing its practical approach.

A core tenet is producing robust software that meets user expectations – providing functional results as designed-

A significant focus centers on efficient resource management; ensuring a successful program from start to finish.

The raw text primarily notes that VB programming limits adaptability compared to pure code sources but the overall objective is effective implementation. \n**Software Engineering – Stages & Processes**

You'll witness distinct stages within software creation:

- **Requirements Gathering:** Understand client expectations and function defined.
- **Design:** Architect the solution, planning all modules' functionalities.

- **Implementation (Coding):** Translate design into actual instructions using programming languages.
- **Testing & Quality Control:** Verify that code's working by running unit test.

The raw text emphasizes an understanding toward efficient solutions rather than relying on a narrow interpretation of the instruction set which has limitations.

Analogies for Clarity

A library's catalogue serves as a helpful example:

Imagine a library – it organizes books based on specific categories. Software engineering is similarly building systems based on design patterns—the core of an organized development project.

As well, consider the process: a recipe in cooking follows precise instructions to give desired final product, software follows similar designs through steps for building functions.

Let's think like puzzle solving.

Key takeaways - focus on planning is important here It is a structured approach, rather than just haphazardly adding elements. Consider these key steps throughout the process:.

- Design: Analyze potential usage.
- Implement: Write code and ensure it's ready to deploy to various units.
- Review: Verify functionality with testing.

Software Engineering strives for practical, reliable results; avoiding missteps.

The key differences are

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Importance of Software

Computer software encompasses practically everything used—from scientific inquiry to everyday commerce. This includes our interactions with technology.

during business operations, transportation & culture. Software fundamentally impacts society as a whole by controlling how we work, interact & discover information.

The key driver for software growth is consistently producing more and better software solutions through careful design.

Software plays an invaluable role in the success and flexibility of systems – it underpins infrastructure from telehealth to military movements - and significantly influences cultural developments. Software also runs the core functionality of operating, powering our devices and connecting various digital elements seamlessly.

The software community is dedicated to improving, simplifying, and lowering costs through continual innovation allowing users across countless applications.